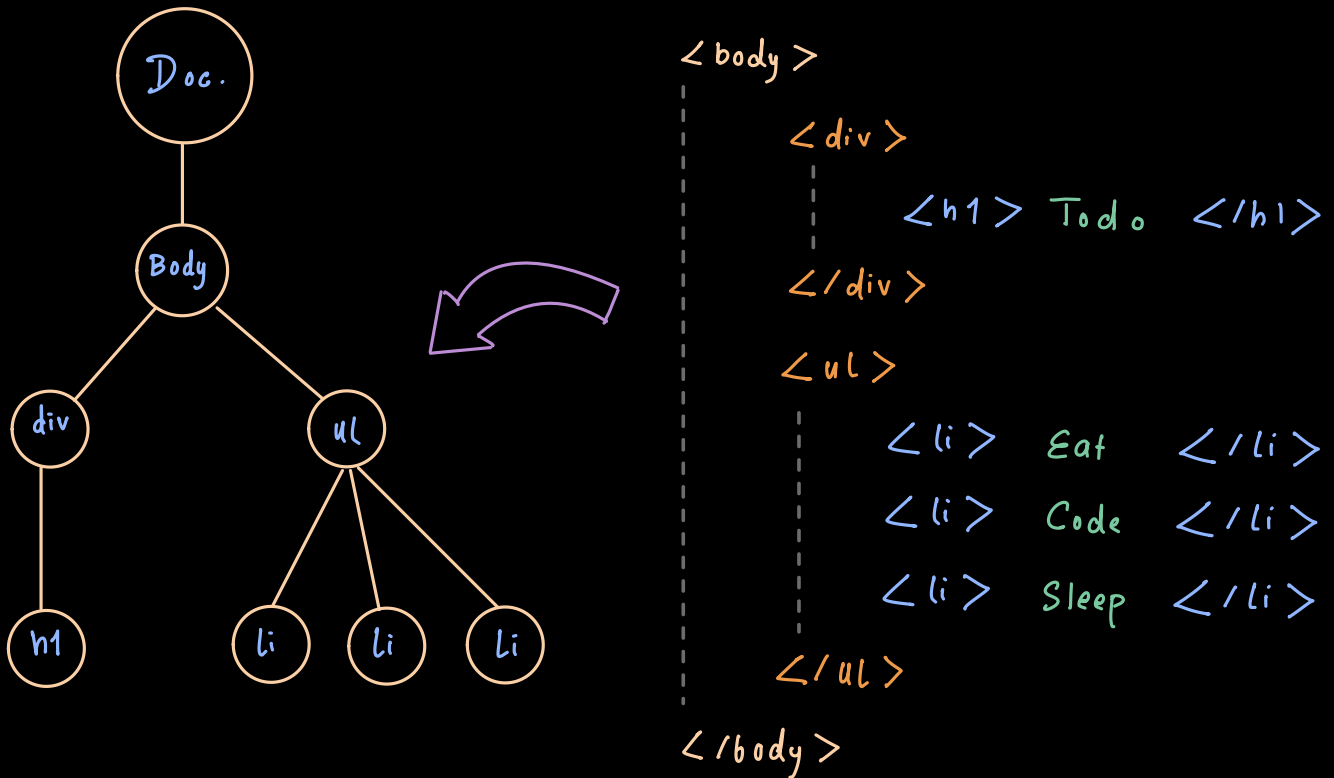


DOM (DOCUMENT OBJECT MODEL)

- Represents a doc. with a logical tree
- It allows us to manipulate / change webpage content (HTML elements)



SELECTING ELEMENT BY ID

getElementById

↳ returns the element as an object or else null, if not found.

`document.getElementById ( mainImg );` ✗

`document.getElementById (" mainImg " );` ✓

↳ gives object.

So we can store it in a value.

```
let imgObj = document.getElementById("mainImg");
```

↳ Now to print actual value :

```
console.dir(imgObj);
```

↳ will give all the individual properties of this object.

To access one property

imgObj.src

imgObj.tagName

imgObj.id

↳ to assign sm else :

```
imgObj.src = " . . . . . " ;
```

SELECTING ELEMENTS BY CLASS NAME

getElementByClassName

↳ returns the elements as an HTML collection or empty collection (if not found)

SELECTING ELEMENTS BY TAG NAME

getElementsByTagName

↳ returns the elements as an HTML collection or empty collection (if not found)

Ex)

```
document.getElementsByTagName("p");
```

QUERY SELECTORS

↳ Allows us to use any CSS selector.

Ex)

```
document.querySelector("p"); // selects first p elements
```

```
document.querySelector("#myId"); // selects first element with  
id = myId
```

```
document.querySelector(".myClass"); // selects first element with  
class = myClass
```

```
document.querySelector>All("p"); // selects all p elements
```

↳ to access all elements

SETTING CONTENT IN OBJECTS

InnerText → Shows visible text contained in a node
Shows whatever is shown on screen

textContent → Shows all the full text
Shows whatever is written in HTML

innerHTML → Shows the full markup.

```
> para.innerText;
< 'Spider-Man is a superhero appearing in American comic books published by Marvel Comics. Created by writer-editor Stan Lee and artist Steve Ditko, he first appeared in the anthology comic book Amazing Fantasy #15 (August 1962) in the Silver Age of Comic Books. He has been featured in comic books, television shows, films, video games, novels, and plays'

> para.textContent;
< '\n      Spider-Man is a superhero appearing in American comic books\n      published by\n      Marvel Comics.\nCreated by writer-editor Stan Lee and artist Steve Ditko, he\n      first appeared in the anthology comic book\nAmazing Fantasy\n      #15 (August 1962) in the Silver Age of Comic Books. He has\n      been featured in comic books, television shows, films, video games,\n      novels, and plays\n      '

> para.innerHTML;
< '\n      <b>Spider-Man</b> is a superhero appearing in American comic books\n      published by\n      <a href="https://www.google.com/search?q=Marvel+Comics">Marvel Comics</a>.\n      Created by writer-editor <b>Stan Lee</b> and artist <b>Steve Ditko</b>, he\n      first appeared in the anthology comic book\n      <a href="https://www.google.com/search?q=Amazing+Fantasy">Amazing Fantasy</a>\n      #15 (August 1962) in the <a href="#">Silver Age of Comic Books</a>. He has\n      been featured in comic books, television shows, films, video games,\n      novels, and plays\n      '
```

For manipulation :

```
para.innerHTML = ' abc ' ;
```

```
para.innerHTML = ' <b> abc </b> ' ;
```

↳ Not bolded

```
para.innerHTML = ' <b> abc </b> ' ;
```

OR

↳ bolded

```
para.innerHTML = ' <b> $ { para.innerHTML } </b> ' ;
```

↳ whatever text
there is bolded.

MANIPULATING ATTRIBUTES

Getters & Setters

```
obj.getAttribute ( attr )
```

```
obj.setAttribute ( attr, val )
```

```
> let img = document.querySelector('img');  
< undefined  
> img  
<   
> img.getAttribute('id');  
< 'spidermanImg'  
> img.setAttribute('id','spidermanImg');  
< undefined
```

MANIPULATING STYLE

↳ style property
↳ obj. style

```
> let header = document.querySelector('h1');  
< undefined  
> header.style;  
< CSSStyleDeclaration {accentColor: '', additiveSymbols: '', alignContent: '', alignItems: '', alignSelf: '', ...}  
> header.style.color;  
< ''  
> header.style.color = 'blue';  
< 'blue'  
> header.style.backgroundColor = 'pink';  
< 'pink'
```

↳ In CSS : background-color
In JS : backgroundColor

Note : We only access inline HTML texts.

Whatever written in CSS is not accessed coz styling is done via CSS exclusively.

CLASSLIST PROPERTY

↳ obj. classList

- 1) `classList.add()` → Add new classes
- 2) `classList.remove()` → Remove classes
- 3) `classList.contains()` → Check if class exists
- 4) `classList.toggle()` → toggle b/w add and remove

↳ If class exists then remove
If class doesn't exist then add

add → TRUE
remove → FALSE

NAVIGATION

- ParentElement
- children
- previousElement / nextElementSibling

```
> let nav = document.querySelector('h4');  
< undefined  
> nav.parentElement  
< ▶ <div class="box"> ... </div>  
> nav.children  
< ▶ HTMLCollection []  
> nav.nextElementSibling  
< ▶ <ul> ... </ul>
```

ADDING ELEMENTS ON PAGE

`document.createElement('p')`

◦ `appendChild(element)`

```
> document.createElement('p');  
< <p></p>  
> let newP = document.createElement('p');  
< undefined  
> let body = document.querySelector('body');  
< undefined  
> body.appendChild(newP);  
< <p></p>
```

- `append(element)` - adds string/text at last
- `prepend(element)` - adds string/text at the beginning
- `insertAdjacent( where, element )`

↳ Refer MDN

REMOVING ELEMENTS FROM PAGE

- `removeChild( element )`
- `remove( element )`

